

Interactive Media 3B | DIGA3009A / DIGA4014A

COURSE SUMMARY

Welcome to Interactive Media 3B — this is an advanced, future-facing course where design, animation, and interactivity are highlighted to shape the next generation of digital experiences. Today, the ability to craft **compelling, interactive**, and **animated** interfaces isn't just a technical asset; it is a sophisticated skill essential for anyone aiming to **engage audiences**, **clarify complex information**, and **communicate visually** in our world which is becoming ever more digital.

Why this course is more important than ever

The web is no longer static. It's not the early-2000s. Audiences demand seamless, responsive, beautiful, and emotionally resonant online experiences. In journalism, business, portfolios, platforms, entertainment, or just day-to-day browsing, your ability to animate, visualise, and interact with data and content, dynamically, is a core expectation of your users. This course equips you with the tools, theory, and applied skills to meet that expectation—blending artistry, technology, and critical thinking.

By the end of this course, you will be well-equipped to tackle the challenges of professional web development environments, armed with both theoretical insights and practical skills.

Independent Learning & Professional Practice

This is a demanding, practical and project-based course. You are expected to research independently, solve technical and design problems, manage your time, and drive your own learning—key habits for success in a creative or tech industry role. Tickets and solutions are included for practice, is ungraded, and is the responsibility of the student to use.

GENERAL TIMETABLE (INSERT DIAGONAL & SEMESTER NO.)

Online	Monday	Async: Technical Lecture Release
3rd Year Lab	Tuesday	08h15 [Design]
3rd Year Lab	Wednesday	10h15 [Professional/Technical]

STAFF

COORDINATOR	LECTURERS	TUTORS
André Gopal [andre.gopal@wits.ac.za]	Imaan Abusamra [imaan.abusamra@wits.ac.za] André Gopal [andre.gopal@wits.ac.za]	Lindokuhle Dlamini [2500080@students.wits.ac.za]; Liam Moodley [2542976@students.wits.ac.za]; Yoav Lipshitz [2567169@students.wits.ac.za]

COURSE OUTCOMES

By the end of this course, students should be able to:

1. **Create meaningful GSAP-driven motion that supports narrative hierarchy, user feedback, spatial transitions, and interaction design rather than decorative effects.** Utilise advanced GSAP features such as timelines, ScrollTrigger, SVG animations, MotionPath, DrawSVG, and Draggable to create engaging interactive experiences.
2. **Optimise animations for performance, responsiveness, and cross-device compatibility.**
3. **Integrate real public APIs** with proper async handling, data rendering, and defensive states while managing technical risks and fallbacks.
4. **Create dynamic, interactive user interfaces** that combine API-fetched data with animated transitions and interactions.
5. **Apply fundamental UI/UX design principles** including usability, accessibility, interactivity, visual hierarchy, responsive design, and animation principles to create user-centered interactive websites.
6. **Demonstrate technical control and code literacy** by writing, debugging, explaining, and modifying vanilla JavaScript, HTML, and CSS, including the ability to critique AI-generated code.
7. **Incorporate accessibility best practices** into animated interactions, ensuring inclusive, accessible designs, and understand the latest in **AI harnessing and control**.
8. Understand professional workflows (Git, documentation, tasks) to deliver a production-ready website that meets defined user needs.
9. **Effectively communicate complex design decisions and technical solutions** clearly and confidently through project presentations and supporting documentation.
10. **Develop and showcase a compelling portfolio-quality project** that demonstrate a combination of technical proficiency, creativity, and thoughtful design—valuable in pursuing internships, employment, or entrepreneurial opportunities.
11. **Gain practical experience with industry-standard tools and libraries** (GSAP, JavaScript, APIs, modern prototyping tools like Figma/XD), preparing for real-world design and development roles in interactive media, development, UX/UI design, creative technology, and digital storytelling.
12. **Clearly articulate the real-world applications of skills learned**, understanding how to leverage these skills in professional interactive media contexts.

ASSIGNMENTS

Name	Description	Date	Weighting
Test 1	HTML/CSS semantics, layout, and responsiveness	5 Aug 2026	13%
Test 2	JavaScript	19 Aug 2026	13%
Test 3	APIs & Data Handling	23 Sep 2026	13%
PRD	Startup Web Site PRD	9 Oct 2026	31%
Exam	Startup Web Site Full Build	TBA	30%
Total	100%		

Assignments are marked according to the Rules and Policies of [University](#), department, and the course (attached on the Ulwazi course page).

CORE READINGS

- **Don't Make Me Think (Revisited)** by Steve Krug — The essential usability bible for intuitive navigation, clear interfaces, and reducing cognitive load. Very good book.
- **The Design of Everyday Things** by Don Norman — Foundational principles of human-centred design, affordances, signifiers, and feedback (highly relevant to interaction and motion design). Used a lot even by academics.
- **Refactoring UI** by Adam Wathan & Steve Schoger — Practical, modern guidance on interface polish, consistency, and visual hierarchy (excellent for design systems and component work).
- **Awwwards.com** — Look here and browse regularly for inspiration for award-winning sites for motion critique examples (Apple, high-end microsites). Most of these sites do not work as everyday websites, they are overly designed. These serve simply as inspiration.
- **100 Things Every Designer Needs to Know About People** by Susan Weinschenk — Psychology-backed insights for better interactions and motion decisions.
- **Apple Human Interface Guidelines** or similar platform guidelines for polish references.
- **MDN Web Docs:** Fetch API, Async/Await, Error Handling, and Performance sections — Authoritative and always up-to-date.
- **Web Performance** guides on MDN (developer.mozilla.org/en-US/docs/Web/Performance) — Critical for your performance plans and avoiding bloat.
- **Articles** on public API best practices, CORS, rate limits, and defensive rendering (search MDN or freeCodeCamp)

CURRICULUM BREAKDOWN	
SEMESTER 2	
Week 1: 20 Jul – 24 Jul	Tuesday: Orientation/Designing with Intention Wednesday: APIs Part 1
Week 2: 27 Jul – 31 Jul	Monday (Async): GSAP Fundamentals Tuesday: Figma Foundations Wednesday: APIs Part 2
Week 3: 3 Aug – 7 Aug	Tuesday: User Research & AI Wednesday: Assignment Test 1 [in-person]
Week 4: 10 Aug – 14 Aug	Monday (Async): GSAP Scrolltrigger Tuesday: Research Analysis Wednesday: Professional Workflows
Week 5: 17 Aug – 21 Aug	Monday (Async): GSAP Animating SVG Tuesday: Independent work session Wednesday: Assignment Test 2
Week 6: 24 Aug – 28 Aug	Monday (Async): API Recap Tuesday: Information Architecture & Storytelling Wednesday: Design Systems
Week 7: 31 Aug – 4 Sep	Monday (Async): GSAP DrawSVG Tuesday: Guest Lecture (UI/UX/Animation) Wednesday: Guest Lecture (UI/UX/Animation)
RESEARCH/STUDY/VACATION BREAK	
28 October – Sunday 01 November (5 days)	
Week 8: 14 Sep – 18 Sep	Monday (Async): GSAP MotionPath Tuesday: Design Systems & Content Design Wednesday: Guest Lecture (UI/UX/Animation)
Week 9: 21 Sep – 25 Sep	Tuesday: Independent work session Wednesday: Assignment 3 Test
Week 10: 28 Sep – 2 Oct	Monday (Async): GSAP Draggable Tuesday: Motion Design Wednesday: AI Literacy Part 1
Week 11: 5 Oct – 9 Oct	Tuesday: Development Ready Designs Wednesday: Independent work session PRD Due: 9 October
Week 12: 12 Oct – 16 Oct	Tuesday: Consultations Wednesday: AI Literacy Part 2
STUDY BREAK	
Wednesday 28 October – Sunday 01 November (5 days)	

EXAMS

Monday 02 November - Friday 27 November